

# Scenario 02

The Clearing Pipeline — Moomoo

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**Role:** Senior Product Manager — Brokerage Clearing and Settlement System, Moomoo Canada

**Domain:** Internal platform · post-trade infrastructure · operational tools

**Stakeholders:** Brokerage Ops · Risk · Compliance · Treasury · Engineering

**Vendors:** DTCC · OCC · CDS · CDCC · Broadridge

**Timeframe:** 6 iterations (6-week quarter)

**Difficulty:** Advanced (matches the JD)

**Target skills:** Platform PM · vendor integration · regulated operations · incident response · post-trade trade lifecycle

**Interview relevance:** This scenario is built specifically for the Moomoo JD "Senior Product Manager — Brokerage Clearing and Settlement System." It leverages Mike's Questrade ops/risk/treasury experience and rehearses the exact tensions this role will face.

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## 1. Setup & Premise

You've been hired as Senior PM for Moomoo Canada's **Clearing and Settlement platform** — the internal post-trade system that handles everything that happens *after* a customer hits "Buy": trade capture, clearing (DTCC/OCC/CDS), netting, settlement (T+1 DVP), reconciliation, corporate actions, regulatory reporting.

Your "customers" are internal: Brokerage Operations, Risk, Treasury, Compliance, and the customer-facing product team who depends on your platform not breaking. Your "users" are Ops analysts who spend all day in dashboards you own.

Moomoo Canada shipped 6 months ago on a mostly-imported platform from HQ (Hong Kong). The platform works for HK's CCASS workflow but was never built for North American clearing. The team has been patching — T+1 settlement workarounds, manual OCC exercise/assignment processing, Broadridge reconciliation in Excel. You're here to turn the patches into a real product.

**Your quarter-ending ask from leadership:** "Get us off operational heroics. Reduce manual break resolution by 50%. Pass the CIRO systems audit in Q3. Don't break anything."

### What's true at iteration 1

- **Daily trade volume:** ~120,000 trades/day across equity + options + FX
- **Break rate:** ~0.8% of trades break → ~960 breaks/day requiring manual analyst intervention
- **Ops team:** 14 analysts spending ~60% of their time on reconciliation
- **US options clearing (OCC):** exercise/assignment processing is 100% manual on expiry days — 500+ accounts/monthly cycle
- **Corporate actions:** handled by a third party with 48-hour latency; customer complaints about dividend timing
- **Broadridge integration:** nightly batch file via SFTP; when it fails, manual re-run required
- **CIRO Systems Audit:** scheduled for Q3; last audit flagged 12 deficiencies around trade surveillance
- **T+1 settlement:** Canada moved to T+1 in May 2024 alongside US; Moomoo's platform is handling it but with 3× the ops workload
- **HK HQ platform team:** responsible for core trading engine; slow to prioritize Canada-specific changes
- **Engineering team:** 4 devs + 1 QA. Two senior, two juniors. One just gave notice.

### Platform constraints (structural)

- HQ controls trade-capture engine; you own post-trade flows downstream
- CIRO systems requirements are non-negotiable — platform must meet Rule 2600 standards
- Regulatory reporting deadlines are hard (CIRO, OSC, FINTRAC); missing them means fines
- DTCC/OCC/CDS have rigid connectivity protocols; vendor change is multi-month
- T+1 settlement means no second chances; fails escalate to DTCC penalties rapidly

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## 2. Win / Lose Conditions

| Outcome          | Condition                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------|
| 🏆 Strong win     | Break rate < 0.4% · Ops capacity freed 30%+ · Audit-ready · Team intact · No settlement fails     |
| ✅ Acceptable win | Break rate < 0.6% · Audit deficiencies addressed · Some ops capacity freed · No major incident    |
| ⚠️ Mixed         | Some improvement but audit deficiencies remain OR team morale cratered OR one settlement incident |
| ❌ Lose           | CIRO audit failure · DTCC penalty · Settlement fail cascade · Key analyst or engineer quits       |

### 3. Stakeholder Personas

#### Darnell (Ops Team Lead)

**Archetype:** Power user (internal) · **Starting state:** ACTIVE (skeptical) · **Voice:** Loud internally, respected  
12 years brokerage ops. Team of 14 analysts. Spends weekends running reconciliations. Has seen three "platform rebuilds" promise the world and deliver pain. Default: skeptical of PMs.

**Wants:** Fewer breaks · auto-matched reconciliations · options expiry automation · getting home for dinner

**Trust signal:** if you solve his actual workflow pain, he becomes your biggest advocate

#### Aisha (Compliance Officer)

**Archetype:** Regulated stakeholder · **Starting state:** ACTIVE · **Voice:** Measured, urgent when triggered  
Owns the CIRO relationship. Preparing the Q3 systems audit. Has veto on any production change affecting trade reporting. Knows Mike's work from past Questrade life.

**Wants:** Clean audit trail · Rule 2600 compliance · trade surveillance fixes · Cease Trade Order (CTO) platform controls

#### Ravi (Risk Manager)

**Archetype:** Peer stakeholder · **Starting state:** ACTIVE · **Voice:** Data-driven, direct

Runs margin and concentration monitoring. Depends on your data feeds being accurate and timely. Has lost trust after 3 data quality incidents.

**Wants:** Real-time exposure data · clean FX positions · reliable options margin feeds · automated margin calls

#### Sophie (Treasury Lead)

**Archetype:** Internal stakeholder · **Starting state:** DORMANT (you haven't met yet) · **Voice:** Quiet, will surface with receipts

Runs cash and collateral management. Feels under-served by your platform. Manual collateral posting to OCC/CDS is eating her team's time.

**Wants:** Auto-collateral optimization · FX reconciliation · intraday liquidity visibility

#### 🚩 Wei (HQ Platform VP)

**Archetype:** Leadership stakeholder · **Starting state:** ACTIVE · **Trust:** 6/10 · **Voice:** Slow, authoritative

Hong Kong based. Owns the global trading platform. Sees Canada as a "small subsidiary." Will push back on any change that diverges from the global platform.

**Wants:** No platform forks · weekly status · global consistency · minimal Canada-specific code

#### 👤 Max (Tech Lead)

**Archetype:** Engineering peer · **Starting state:** ACTIVE · **Voice:** Pragmatic, honest

Senior engineer on your team. 8 years experience, ex-CIBC. Understands settlement plumbing better than most PMs. Frustrated by HQ platform constraints.

**Wants:** Tech debt paid down · clean APIs with HQ · room to build vs patch

#### 🤝 OCC / DTCC / CDS Account Managers (external vendors)

**Archetype:** Vendor stakeholders · **Starting state:** ACTIVE · **Voice:** Formal

Your vendor contacts at the three major clearinghouses. Each has their own release schedules, integration patterns, and politics.

**Want:** Timely responses to their inquiries · clean data submissions · early notice of platform changes

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## 4. Product Backlog

Format: effort value who benefits

### Ops / operational tooling

#### Auto-Break Resolution Engine (Phase 1 — low-complexity breaks)

effort: 10 value: very high Darnell · Auto-resolves ~60% of breaks (cash/stock matches within tolerance). Frees 30% of ops capacity.

#### OCC Exercise/Assignment Automation

effort: 12 value: very high Darnell · Ravi · Replaces manual monthly-expiry process for 500+ accounts. Requires OCC integration work.

#### Settlement Break Dashboard (real-time, drill-down)

effort: 6 value: high Darnell · Replaces Darnell's morning 3-spreadsheet ritual. Lifts analyst productivity 15%.

#### Corporate Actions Workflow Integration

effort: 14   value: high   Darnell · Ravi   · Replaces third-party with in-platform. Eliminates 48-hr latency on dividends and mergers.

### Broadridge Reconciliation Automation

effort: 7   value: medium-high   Darnell   · Removes nightly manual re-runs. Builds robustness into the SFTP feed.

### Trade Break Root-Cause Tagging

effort: 3   value: medium   Darnell · Max   · Adds taxonomy so breaks are tagged at resolution. Enables Phase 2 automation prioritization.

## Risk & compliance

### CIRO Audit Preparation — Trade Surveillance Module

effort: 11   value: critical   Aisha   · Closes 8 of 12 audit deficiencies. Non-negotiable for Q3 audit.

### CTO (Cease Trade Order) Enforcement Pipeline

effort: 8   value: high   Aisha · Darnell   · Automates platform-level restrictions on CTO'd securities. Currently manual via spreadsheet; high error risk.

### Regulatory Reporting Automation (CIRO + OSC + FINTRAC)

effort: 10   value: high   Aisha   · Replaces scripts maintained by one junior dev. Reduces miss risk.

### Margin Call Automation

effort: 9   value: high   Ravi · Darnell   · Replaces daily manual margin call issuance. Connects to auto-sellout on breach.

### Real-time Exposure Feed to Risk Team

effort: 5   value: high   Ravi   · Replaces end-of-day batch with real-time. Required for HFT fraud detection.

## Treasury

### Auto Collateral Optimizer (OCC / CDS)

effort: 8   value: medium-high   Sophie   · Algorithm-driven collateral posting to minimize cash-on-deposit. Saves ~\$500K/yr interest on collateral.

### FX Reconciliation Module

effort: 5   value: medium   Sophie   · Replaces manual FX book rec (currently in Excel).

## Tech / platform

### Event-Driven Settlement Architecture

effort: 15   benefit: strategic   · Major rebuild from batch → event-driven. Reduces future break rate by 60% and unlocks real-time features. Multi-quarter investment.

### HQ Platform API Contract

effort: 6 benefit · Formalizes integration with HQ's trading engine. Reduces HQ-related incidents 40%. Political cost with Wei.

### Observability Platform for Post-Trade

effort: 4 benefit · End-to-end trace of every trade. Cuts incident triage time 70%.

### Data Quality Monitoring

effort: 3 benefit · Catches feed corruption before it hits Risk. Reduces Ravi's trust deficit.

### Disaster Recovery Runbook + Drill

effort: 4 benefit · Required by CIRO audit. Also shows up in the Risk audit. Nobody wants to do this.

## 5. Event Deck (20 cards)

### Stakeholder events

#### STAKEHOLDER · FORCED ITER 2

##### Wei (HQ) wants you to "align" with global platform

"We don't understand why Canada needs a separate break-resolution engine. Our HK team has already built this. Use ours." Problem: HK's engine doesn't understand DTCC or OCC clearing cycles.

- **Accept and fork** — start using HK's; plan patches for DTCC integration. Save 10 pts; add 40 pts of future patch work. HQ trust +2.
- **Propose parallel development with shared interfaces** — show the diff with data. Get buy-in for Canada-specific version. HQ trust neutral; Max morale +1.
- **Escalate to your CEO** — this is platform autonomy. Nuclear option. High cost.

#### STAKEHOLDER · WEIGHTED

##### Darnell invites you to Ops weekly

Darnell's team does a Monday 7am meeting to review weekend breaks. He offers you a standing invite. "If you're really serious about ops tooling."

- **Attend weekly for the quarter** — time cost 1 hr/week. Darnell trust +5. Ops insights compound.
- **Attend monthly + send deputy** — symbolic. Darnell trust +2.
- **Decline politely, schedule 1:1s instead** — misses the in-flight signal. Darnell trust -1.

#### STAKEHOLDER · WEIGHTED BY BEHAVIOR

##### Aisha: "I need 3 days of dev time for audit prep"

Aisha lands in the middle of a sprint. Needs documentation + data extracts for CIRO audit. Your devs are mid-sprint.

- **Prioritize her fully** — drop mid-sprint work. Velocity -5. Aisha trust +2. Audit risk reduced.
- **Co-plan with her** — delay 1 iteration; she gets it. Aisha trust 0. No velocity hit.

- **Delegate to junior dev alone** — cheap but risky. Aisha notices absence of senior attention.

#### STAKEHOLDER · RANDOM

##### Sophie (Treasury) finally introduces herself

You've been in-role 8 weeks. Sophie sends a calm email: "I don't think you have Treasury on your roadmap. I'd like to understand your vision." First signal from a dormant stakeholder.

- **Schedule deep discovery (2 hrs)** — investment that pays back. Sophie state: DORMANT → INTERESTED.
- **Defer to iteration 4** — she feels deprioritized. State stays DORMANT.
- **Add Treasury items to backlog without talking** — guesses at what she wants. Medium risk.

## Operational events

#### OPS · HIGH IMPACT · RANDOM

##### Options expiry Friday: 60+ OCC exercise breaks

A Friday expiry. Ops team pulls an all-nighter. Clients wake up Saturday confused. Darnell is furious.

*Forces you into an incident. Measures your response speed and ability to triage.*

- **Take command of the incident** — lead comms, coordinate ops + risk + engineering. Morale +2.
- **Delegate to Max and handle stakeholder comms** — you own customer-facing narrative; Max fixes.
- **Stay out; let ops handle** — you're not technical enough. Darnell trust -2.

#### OPS · WEIGHTED

##### DTCC settlement fail notice

Moomoo fails to settle \$2.3M of trades on a T+1 cycle. DTCC issues a formal fail notice. If this happens 3x in a quarter, DTCC imposes penalties.

- **Immediate post-mortem + root cause fix** — 1 sprint dedicated. Tech debt -10. Customer-facing features stall.
- **Incident report to CIRO + quick patch** — keep velocity moving. Risk of repeat.
- **Patch silently** — non-disclosure risk if regulator asks. High risk move.

#### REGULATORY · FORCED ITER 5

##### CIRO Audit Pre-Assessment Visit

Two CIRO auditors arrive for a scoping visit. They review your state of readiness for Q3 audit.

*Outcome depends on earlier decisions: if you invested in Trade Surveillance module (iter 1-3), they're satisfied. If you didn't, they flag additional deficiencies.*

## Team events

#### TEAM · FORCED ITER 1

##### Your senior dev gives notice (Goodbye Sprint)

Rajesh tells you Monday: "4 weeks notice. Accepted a role at RBC Capital Markets." He owns the OCC integration — nobody else understands it.

- **Pair him with Max on documentation for 2 iterations** — pair programming to transfer knowledge. Velocity -3.
- **Get him to ship one last big thing** — accept the knowledge transfer gap. Risk: OCC system is a black box after he leaves.
- **Try to counter-offer** — political, needs Wei's approval. Rarely works.

#### TEAM · WEIGHTED

##### Max (tech lead) proposes the event-driven architecture rebuild

Max pitches a major rebuild: batch → event-driven. Would take 2 quarters. Would transform everything. Risky.

- **Green-light a 2-week spike** — investigate feasibility. Safe learning.
- **Commit to a phased rebuild** — Max gets dream project. Customer-facing work slows.
- **Defer, focus on incremental wins** — Max morale -1; pragmatic.

#### TEAM · WEIGHTED

##### Junior dev makes a production bug at worst time

Arjun (new junior) deploys a change during CIRO audit prep week. Small bug but visible to auditors.

- **Fix publicly and own it** — take blame upward. Team morale +3. Trust with Aisha +1.
- **Fix quietly** — minimize visibility. Team learns nothing.
- **Blame protocol** — unclear process. Team morale -3.

## Vendor events

#### VENDOR · RANDOM

##### OCC releases protocol update — breaking change in 45 days

OCC announces a required protocol update. Non-optional. Affects exercise/assignment flow.

- **Prioritize immediately** — absorb 6 pts into next iteration. Painful but safe.
- **Request extension** — OCC rarely grants. Risk.
- **Collaborate with Ops on manual workaround** — buy time, risky.

#### VENDOR · RANDOM

##### Broadridge reconciliation file corruption

Friday night: the nightly reconciliation feed has bad data. Ops will discover Monday morning unless handled.

## Strategic events

#### STRATEGIC · RANDOM ITER 3

##### CEO wants to know your 12-month platform vision

Moomoo Canada CEO schedules a 30-min meeting. "Paint me the picture: what's the platform look like in 12 months?" Your answer shapes next year's budget.

- **Ambitious vision (event-driven rebuild)** — budget ask: \$3M. High risk, high reward.
- **Pragmatic vision (incremental fixes + audit compliance)** — budget ask: \$1M. Low risk.
- **Ask for time to prepare** — buy 2 weeks to bring data. Risk: CEO loses patience.

#### STRATEGIC · RANDOM

##### Customer-facing Product team wants cryptocurrency integration

Moomoo's customer-facing team (different org) announces crypto feature. Your platform doesn't handle crypto custody or settlement. You'd own the post-trade side.

## 6. Scenario Narrative Beats

| Iter | Framing                                                                                                                                                                                        |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | "Week 1. Rajesh told you he's leaving. Darnell's team had 4 breaks over the weekend. Aisha wants an audit prep meeting. Wei forwards you a global platform rollout plan. Where do you invest?" |
| 2    | "Week 2. Wei is pushing harder on platform alignment. You've had no contact with Treasury. Your options expiry process is still 100% manual."                                                  |
| 3    | "Week 3. CEO wants your vision. Two more breaks surfaced. Ravi's team reported data-quality incident #4. Quarterly review is 3 weeks out."                                                     |
| 4    | "Week 4 — halfway. Audit pre-assessment is next week. Some of your iter-1 decisions are showing up as good or bad. You can still course-correct."                                              |
| 5    | "Week 5. Audit pre-assessment happened. CIRO noticed. Options expiry is Friday. Rajesh leaves next Friday."                                                                                    |
| 6    | "Week 6. Final sprint. Whatever's not ready is not ready. Darnell offered to write a testimonial if you shipped the break dashboard. Audit is 4 weeks away."                                   |

## 7. Interview Skills This Scenario Rehearses

Moomoo's "Senior PM — Clearing & Settlement" interview will focus on:

- **Platform PM mindset** — internal "customers" (Ops, Risk, Compliance) differ from consumer PM
- **Regulatory sophistication** — CIRO, DTCC, OCC fluency is table stakes; show depth
- **Vendor/integration management** — explaining a DTCC change to a dev team
- **Incident response** — how you lead during settlement failures or audit findings
- **Stakeholder navigation** — especially HQ (HK) ↔ local (Canada) tension
- **Trade-off thinking** — infrastructure investment vs incremental fixes
- **Domain storytelling** — fluency with trade lifecycle, post-trade flows, settlement cycles (Mike has this at Questrade)

Every event in this scenario maps to a question type you'll face. Run it 3x, build reflexes, use the best decisions as behavioral interview stories.

**Why this scenario works for you:** you have lived every role in the stakeholder persona list. You know Darnell (you were Darnell). You know Aisha (you worked with compliance). You know the Rajesh problem (you mentor analysts). You know Wei (you worked with global product owners). This scenario converts your 13 years of experience into interview narratives.